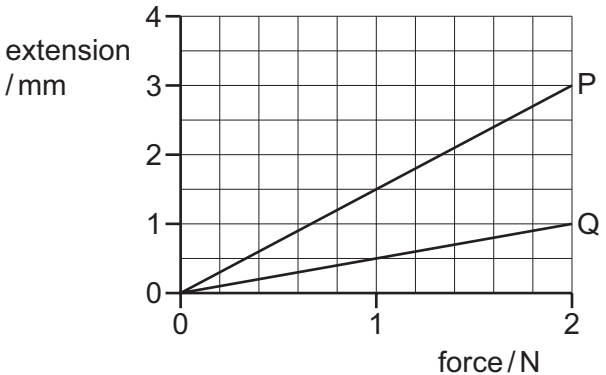
19 Two wires, P and Q, made of the same material, are stretched with an increasing force.

A graph is plotted of the variation with force of the extension of each wire.



The wires have the same original length but different diameters.

What is the ratio $\frac{\text{diameter of wire Q}}{\text{diameter of wire P}}$?

A $\frac{1}{3}$ **B** $\frac{1}{\sqrt{3}}$ **C** $\sqrt{3}$ **D** 3